Data Mining Final Project: Student Academic Performance Analysis

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Dataset: Students Performance Dataset (Kaggle)

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1. Introduction

The objective of this project is to apply data mining techniques to a comprehensive educational dataset to identify the primary drivers of student success. By analyzing over 5,000 student records, the project aims to bridge the gap between lifestyle choices and academic outcomes. This analysis provides a framework for predicting grades and segmenting student behavior into actionable categories.

2. Dataset Overview

The project utilizes the "Students Performance Dataset" sourced from Kaggle.

- **Scale:** The dataset contains 5,000+ individual student records.
- **Dimensions:** It originally included 22 features covering academic and personal data points.
- **Target Variable:** The primary target is the Grade Category, ranging from A to F.
- **Key Features:** Variables include Attendance %, Midterm and Final Scores, Assignments and Quizzes Averages, Participation, Study Hours per Week, Sleep Hours, Stress Levels, Family Income, and Parent Education.

3. Data Preprocessing (Homework 3)

To prepare the data for modeling, several preprocessing techniques were implemented to ensure data quality and model performance:

3.1 Data Reduction

Non-predictive features, such as Student IDs, Names, and Emails, were removed. This dimensionality reduction improved model efficiency by focusing on the 18 most relevant features.

3.2 Data Transformation & Cleaning

- **Label Encoding:** Categorical variables were converted into numerical formats to allow for mathematical processing by machine learning algorithms.

- **Standardization:** Feature scaling (StandardScaler) was applied particularly for the K-Means clustering algorithm to ensure that features like "Study Hours" and "Total Score" were weighted equally.
- **Data Splitting:** The dataset was partitioned into a training set (80%) and a testing set (20%) to validate model accuracy.

4. Exploratory Data Analysis (EDA)

EDA was conducted to visualize trends before applying machine learning models:

- **Grade Distribution:** A count plot showed that the majority of students fall into the 'C' and 'D' grade categories.
- **Correlation Matrix:** A heatmap revealed that Midterm and Final scores have the strongest correlation with the Total Score, while factors like Attendance and Study Hours also show measurable positive impacts.
- **Relationship Analysis:** Scatter plots between Study Hours and Total Score confirmed that while more study time generally leads to better grades, there is significant variance influenced by other lifestyle factors.

5. Methodology and Modeling (Homework 4)

5.1 Cluster Analysis (Unsupervised Learning)

Using **K-Means Clustering** with $k=3$, students were segmented into three distinct behavioral groups:

- **Cluster 0 (High Achievers):** Characterized by high study hours and top academic scores.
- **Cluster 1 (Medium Performers):** Characterized by average effort and mid-range scores.
- **Cluster 2 (Low Performers):** Characterized by low study hours and lower overall performance.

5.2 Classification and Regression (Supervised Learning)

- **Random Forest Classifier:** This model was used to predict the discrete Grade category (A–F) and identify feature importance.
- **Linear Regression:** Applied to predict the numerical Total Score. The model achieved a high R^2 score on the test set, indicating that the predicted scores closely match the actual academic performance.

6. Key Insights and Conclusion

The project yielded several actionable insights for educational stakeholders:

- **Attendance & Effort:** Attendance and Study Hours remain the most reliable predictors of student success.
- **Lifestyle Impact:** Stress levels and sleep hours have a quantifiable effect on academic outcomes, suggesting that student well-being is critical for performance.
- **Model Success:** The combination of clustering and regression provided a robust method for identifying at-risk students and predicting final outcomes with high accuracy.

Tools & Technologies: Python, Pandas, NumPy, Scikit-Learn, Seaborn, and Matplotlib.